

SMM634 Analytics Methods for Business

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Individual Assignment

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1 Model Selection for Mortality

1.1 Study Aim and Variable Selection Rationale

This study examines in-hospital mortality among patients in a Virginia medical center, focusing on demographics, physiological indicators, and clinical assessments. In-hospital mortality serves as a measure of healthcare quality, guiding resource allocation and identifying factors linked to patient deaths.

Rationale for Variable Selection:

- **Age and Gender:** Older age and male gender may increase mortality risks due to physiological decline or health disparities.
- **Body Mass Index (BMI):** Extremes like underweight or obesity may reflect poor health or nutrition, increasing mortality risk.
- **Severity and Risk Scores:** These provide measures of a patient's condition and risk, offering predictive value for mortality.
- **Vital Signs (SpO2, SBP, DBP, Pulse, Respiratory Rate, Temperature):** Abnormal values may signal acute crises or systemic failure linked to mortality.
- **AVPU Score:** Lower scores reflect severe neurological compromise, strongly associated with mortality.
- **Length of Stay (los):** Long stays may indicate serious health issues, while short stays before death suggest rapid decline.

Logistic regression was chosen to model the probability of mortality. The initial model (Equation 1) included predictors with high p-values (Table 1) and was refined using stepwise AIC.

$$\widehat{\text{died}}_i \sim \text{Bernoulli}(\widehat{p}_i)$$

$$\begin{aligned} \log \left(\frac{\hat{p}_i}{1 - \hat{p}_i} \right) = & \hat{\beta}_0 + \hat{\beta}_1(\text{age}) + \hat{\beta}_2(\text{gender: Male}) + \hat{\beta}_3(\text{BMI}) \\ & + \hat{\beta}_4(\text{severity: Moderate}) + \hat{\beta}_5(\text{severity: Major}) + \hat{\beta}_6(\text{severity: Extreme}) \\ & + \hat{\beta}_7(\text{risk: Moderate}) + \hat{\beta}_8(\text{risk: Major}) + \hat{\beta}_9(\text{risk: Extreme}) \\ & + \hat{\beta}_{10}(\text{SpO}_2) + \hat{\beta}_{11}(\text{SBP}) + \hat{\beta}_{12}(\text{DBP}) \\ & + \hat{\beta}_{13}(\text{pulse}) + \hat{\beta}_{14}(\text{respiratory rate}) + \hat{\beta}_{15}(\text{AVPU: Verbal}) \\ & + \hat{\beta}_{16}(\text{AVPU: Pain}) + \hat{\beta}_{17}(\text{AVPU: Unresponsive}) + \hat{\beta}_{18}(\text{temperature}) \quad (1) \end{aligned}$$

Table 1: Coefficients of the Logistic Regression Model

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	13.557752	19.496831	0.695	0.48682	
los	-0.145508	0.052669	-2.763	0.00573	**
age	0.029573	0.017656	1.675	0.09393	.
genderM	0.696642	0.473986	1.470	0.14163	
bmi	-0.062184	0.041287	-1.506	0.13203	
severity2 - Moderate	-0.084054	1.289244	-0.065	0.94802	
severity3 - Major	1.355750	1.394645	0.972	0.33100	
severity4 - Extreme	2.187927	1.526761	1.433	0.15184	
spO2	-0.107721	0.057522	-1.873	0.06111	.
sbp	-0.004828	0.011477	-0.421	0.67400	
dbp	0.009121	0.017376	0.525	0.59962	
pulse	0.004836	0.012157	0.398	0.69078	
respiratory	0.066276	0.047633	1.391	0.16411	
avpuP	1.081105	1.302194	0.830	0.40642	
avpuU	1.701291	1.031009	1.650	0.09892	.
avpuV	-1.408210	1.295911	-1.087	0.27719	
temp	-0.103986	0.182804	-0.569	0.56947	
risk2 - Moderate	-0.597477	1.150018	-0.520	0.60339	
risk3 - Major	0.595140	1.161712	0.512	0.60844	
risk4 - Extreme	2.482343	1.250831	1.985	0.04719	*

Significance codes: 0 ‘***’, 0.001 ‘**’, 0.01 ‘*’, 0.05 ‘.’, 0.1 ‘ ’

GAM was used to check for uncaptured non-linear relationships:

$$\widehat{\text{died}}_i \sim \text{Bernoulli}(\widehat{p}_i)$$

$$\begin{aligned}
\log\left(\frac{\hat{p}_i}{1 - \hat{p}_i}\right) = & \hat{\beta}_0 + \hat{\beta}_1(\text{gender: Male}) \\
& + \hat{\beta}_2(\text{AVPU: Pain}) + \hat{\beta}_3(\text{AVPU: Verbal}) \\
& + \hat{\beta}_4(\text{AVPU: Unresponsive}) + \hat{\beta}_5(\text{risk: Moderate}) \\
& + \hat{\beta}_6(\text{risk: Major}) + \hat{\beta}_7(\text{risk: Extreme}) \\
& + \hat{s}(\text{los}) + \hat{s}(\text{age}) + \hat{s}(\text{BMI}) \\
& + \hat{s}(\text{respiratory rate}) + \hat{s}(\text{spO2})
\end{aligned} \tag{2}$$

The EDF for $s(\text{los})$ is greater than 1 (EDF = 2.309) indicating that the relationship between length of stay (**los**) and the outcome (**died**) is slightly non-linear (Table 2).

Table 2: Significance of Smooth Terms in GAM

Predictor	EDF	Chi-Sq	p-value	Interpretation
$s(\text{los})$	2.309	13.916	0.00461	Significant and non-linear
$s(\text{age})$	1.001	1.768	0.18361	Linear and not significant
$s(\text{BMI})$	1.001	1.542	0.21436	Linear and not significant
$s(\text{respiratory})$	1.000	2.594	0.10736	Linear and marginally non-significant
$s(\text{spO2})$	1.287	4.070	0.04366	Significant and slightly non-linear

The smooth term plot for **los** shows high variability in confidence intervals, due to limited observations (Figure 1). The EDF of $s(\text{los}) = 2.3$ suggests slight non-linearity but lacks practical significance. Thus, it is reasonable to revert to a simpler model with **los** as a linear term.

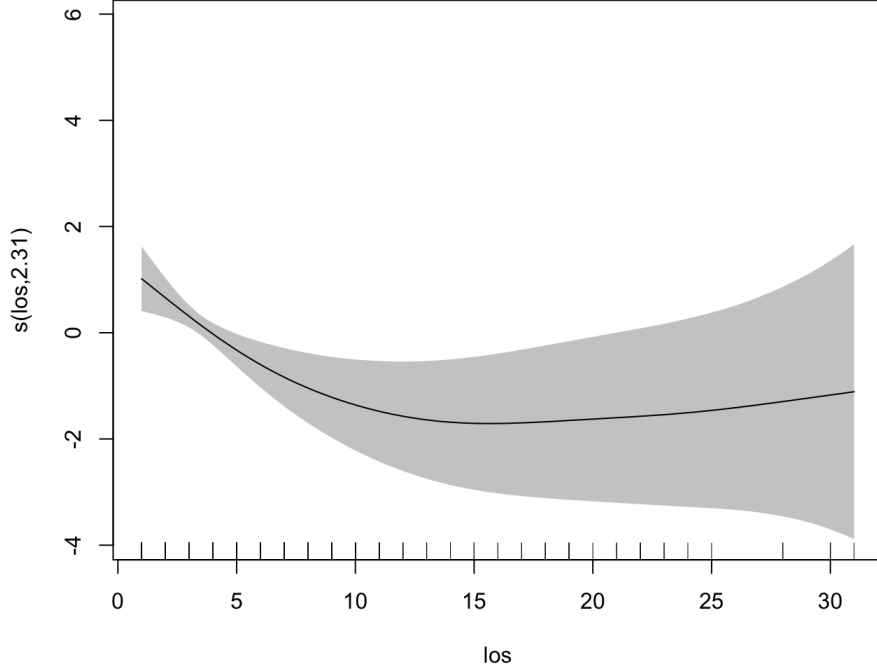


Figure 1: GAM `los` model

Three models tested interactions between variables: `age` with `severity`, BMI with `risk`, and `age` with `risk`. None of the interaction terms were significant. Including these interactions added complexity, the simpler stepwise AIC model remains the best choice. The multicollinearity of the variables was assessed, all VIF values indicated no significant multicollinearity issues among the predictors.

It is important to evaluate the model's performance:

$$died_i \sim \text{Bernoulli}(\hat{p}_i)$$

$$\begin{aligned} \log \left(\frac{\hat{p}_i}{1 - \hat{p}_i} \right) = & \hat{\beta}_0 + \hat{\beta}_1(\text{los}) + \hat{\beta}_2(\text{age}) + \hat{\beta}_3(\text{gender: Male}) \\ & + \hat{\beta}_4(\text{BMI}) + \hat{\beta}_5(\text{spO2}) + \hat{\beta}_6(\text{respiratory}) + \hat{\beta}_7(\text{AVPU: Pain}) \\ & + \hat{\beta}_8(\text{AVPU: Verbal}) + \hat{\beta}_9(\text{AVPU: Unresponsive}) + \hat{\beta}_{10}(\text{risk: Moderate}) \\ & + \hat{\beta}_{11}(\text{risk: Major}) + \hat{\beta}_{12}(\text{risk: Extreme}) \end{aligned} \quad (3)$$

With a threshold of 0.5, the model achieves high accuracy (97.4%) and a low false positive rate (0.4%) but a low true positive rate (36.4%), missing many deaths (Table 3). For better

detection of deaths, the optimal threshold of 0.026, identified from the ROC curve, increases sensitivity to 84.8% but raises the false positive rate to 17.1% and lowers accuracy to 82.9% (Table 4).

Table 3: Confusion Matrix and Metrics for Threshold = 0.5

		Actual		Metric	Value
		0	1	Accuracy	0.974
Predicted	0	941	21	True Positive Rate (Sensitivity)	0.364
	1	4	12	False Positive Rate	0.004

Table 4: Confusion Matrix and Metrics for Optimal Threshold

		Actual		Metric	Value
		0	1	Accuracy	0.829
Predicted	0	783	5	True Positive Rate (Sensitivity)	0.848
	1	162	28	False Positive Rate	0.171

The ROC curve closer to the top-left corner indicates good performance, and the AUC of 0.913 confirms this.

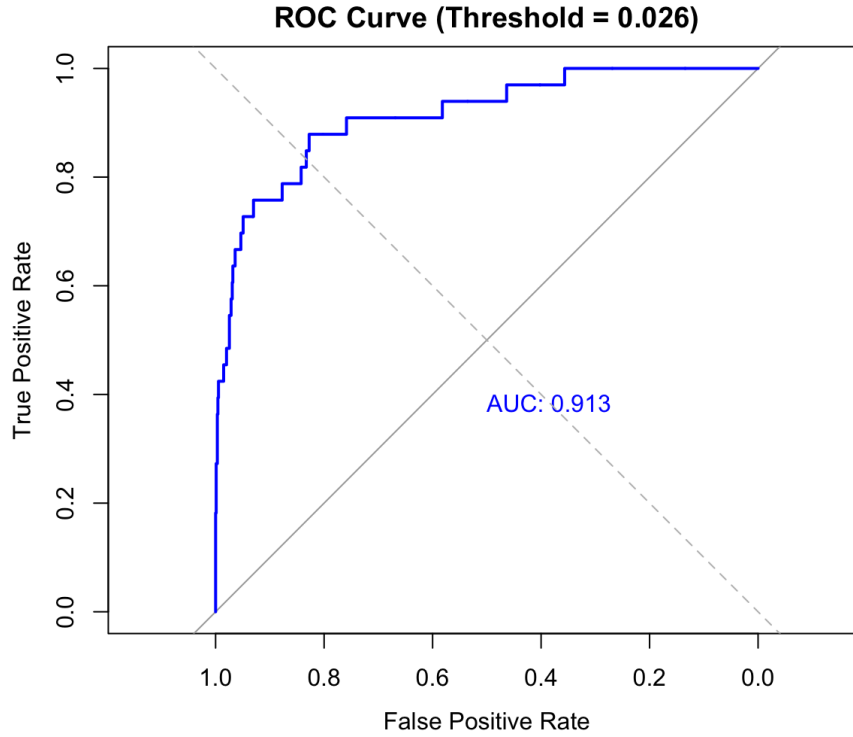


Figure 2: ROC Curve, showing strong predictive ability with AUC = 0.913. The dashed line represents random guessing.

Therefore, the chosen logistic regression model can be represented as:

$$died_i \sim \text{Bernoulli}(p_i)$$

$$\begin{aligned} \log \left(\frac{p_i}{1 - p_i} \right) = & \beta_0 + \beta_1(\text{los}) + \beta_2(\text{age}) + \beta_3(\text{gender: Male}) \\ & + \beta_4(\text{BMI}) + \beta_5(\text{spO2}) + \beta_6(\text{respiratory}) + \beta_7(\text{AVPU: Pain}) \\ & + \beta_8(\text{AVPU: Verbal}) + \beta_9(\text{AVPU: Unresponsive}) + \beta_{10}(\text{risk: Moderate}) \\ & + \beta_{11}(\text{risk: Major}) + \beta_{12}(\text{risk: Extreme}) \end{aligned} \quad (4)$$

2 Model Selection for Length of Stay

2.1 Study Aim and Variable Selection Rationale

The `los` model variables were selected similarly to the mortality model, as demographics, vital signs, and clinical assessments influence both mortality and hospitalization length.

2.2 Model justification

The response variable (`los`) has a mean (`mean_los` = 4.69) and variance (`var_los` = 18.02), indicating overdispersion and violating the Poisson assumption. To address this, the Negative Binomial distribution was chosen for its ability to model variance independently of the mean, making it suitable for highly overdispersed data like `los`.

The initial model identifies **died**, **gender**, and **severity** as significant predictors (Table 5). However, focusing only on z-values may exclude variables that enhance overall model fit when combined with others.

Table 5: Coefficients of the Initial Negative Binomial Model

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	-3.1868660	2.4284529	-1.312	0.18942	
died	-0.5937305	0.1297386	-4.576	4.73e-06	***
age	0.0019088	0.0015748	1.212	0.22548	
genderM	-0.1318880	0.0448635	-2.940	0.00328	**
bmi	0.0006506	0.0033551	0.194	0.84626	
severity2 - Moderate	0.2691877	0.0652641	4.125	3.71e-05	***
severity3 - Major	0.6658413	0.0832707	7.996	1.28e-15	***
severity4 - Extreme	1.2142324	0.1308613	9.279	< 2e - 16	***
sp02	0.0096872	0.0072544	1.335	0.18176	
sbp	-0.0006259	0.0011569	-0.541	0.58851	
dbp	-0.0022799	0.0018821	-1.211	0.22577	
pulse	0.0017044	0.0012912	1.320	0.18682	
respiratory	0.0017148	0.0073502	0.233	0.81553	
avpuP	0.0531253	0.2119884	0.251	0.80212	
avpuU	-0.0786119	0.1895718	-0.415	0.67838	
avpuV	-0.1877853	0.1548152	-1.213	0.22514	
temp	0.0337767	0.0230657	1.464	0.14309	
risk2 - Moderate	0.0100724	0.0612525	0.164	0.86938	
risk3 - Major	0.0357635	0.0819451	0.436	0.66252	
risk4 - Extreme	0.2388979	0.1327681	1.799	0.07196	.

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' '.

AIC: 4651.9

Stepwise AIC identified `died`, `gender`, `severity`, `dbp` (diastolic blood pressure), and `temp` (temperature) as optimal predictors (Table 6). While `dbp` and `temp` were not individually significant, their inclusion improved AIC and reflects their clinical relevance to patient health and hospital stay duration.

Table 6: Coefficients of the Step AIC Negative Binomial Model

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	-2.406971	2.128622	-1.131	0.25815	
died	-0.531563	0.124798	-4.259	2.05e-05	***
genderM	-0.124375	0.044349	-2.804	0.00504	**
severity2 - Moderate	0.281240	0.062386	4.508	6.54e-06	***
severity3 - Major	0.723828	0.065887	10.986	<2e-16	***
severity4 - Extreme	1.390811	0.088464	15.722	<2e-16	***
dbp	-0.002537	0.001521	-1.668	0.09540	.
temp	0.037800	0.021704	1.742	0.08157	.

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' '.

AIC: 4638.

Generalized Additive Model (GAM) tested non-linear relationships in `dbp` and `temp`. The EDF for `s(dbp)` and `s(temp)` were near 1 (Table 7), indicating no significant non-linear effects. The stepwise AIC model was retained.

Table 7: Parametric Coefficients and Smooth Terms

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	1.11229	0.05460	20.371	<2e-16	***
died	-0.53165	0.12530	-4.243	2.21e-05	***
genderM	-0.12441	0.04452	-2.795	0.0052	**
severity2 - Moderate	0.28123	0.06258	4.494	6.99e-06	***
severity3 - Major	0.72379	0.06611	10.948	<2e-16	***
severity4 - Extreme	1.39087	0.08886	15.653	<2e-16	***
Smooth Term	edf	Ref.df	Chi.sq	p-value	Significance
s(dbp)	1.001	1.002	2.776	0.0959	.
s(temp)	1.001	1.002	3.009	0.0830	.

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' '.

Several interaction terms were tested to see if relationships between variables and `los` depended on other factors. `severity` with `gender`, `temperature` with `mortality`, and BMI with `gender` were explored to check for combined effects. Most interactions, were insignificant. Including these terms added complexity. The simpler stepwise AIC model with only main effects performed equally well and is easier to interpret (Equation 5).

$$\begin{aligned} \widehat{\text{los}}_i &\sim \text{Negative Binomial}(\widehat{\mu}_i, \widehat{\theta}) \\ \log(\widehat{\mu}_i) &= \widehat{\beta}_0 + \widehat{\beta}_1(\text{died}) + \widehat{\beta}_2(\text{gender: Male}) \\ &\quad + \widehat{\beta}_3(\text{severity: Moderate}) + \widehat{\beta}_4(\text{severity: Major}) \\ &\quad + \widehat{\beta}_5(\text{severity: Extreme}) + \widehat{\beta}_6(\text{dbp}) \\ &\quad + \widehat{\beta}_7(\text{temp}) \end{aligned} \tag{5}$$

It is important to check the residuals to ensure the model assumptions are met, such as the appropriateness of the negative binomial distribution, absence of systematic patterns, and adequate handling of overdispersion.

Residuals vs Fitted Plot. Slight heteroscedasticity is observed, possibly due to unmeasured variables or overdispersion in the data. Missing covariates, such as comorbidities or treatment variations, may contribute to this variability.

Q-Q Plot. The residuals deviate from theoretical quantiles in the tails, suggesting that while the negative binomial model addresses overdispersion, it does not fully capture extreme

values (e.g., outliers like 6090 and 719). These deviations are common and might indicate rare patient circumstances.

Scale-Location Plot. The non-uniform spread of residuals suggests heteroscedasticity, where variance increases with fitted values. Neither GAM results ($\text{EDF} = 1$) nor interaction terms (e.g., `severity:gender`, `age:gender`) explain this issue. This pattern likely stems from overdispersion in count data.

Residuals vs Leverage Plot. Influential points (e.g., 609, 166, 719) exert high leverage. As these are valid data points, they may reflect unmeasured factors such as rare conditions or treatments.

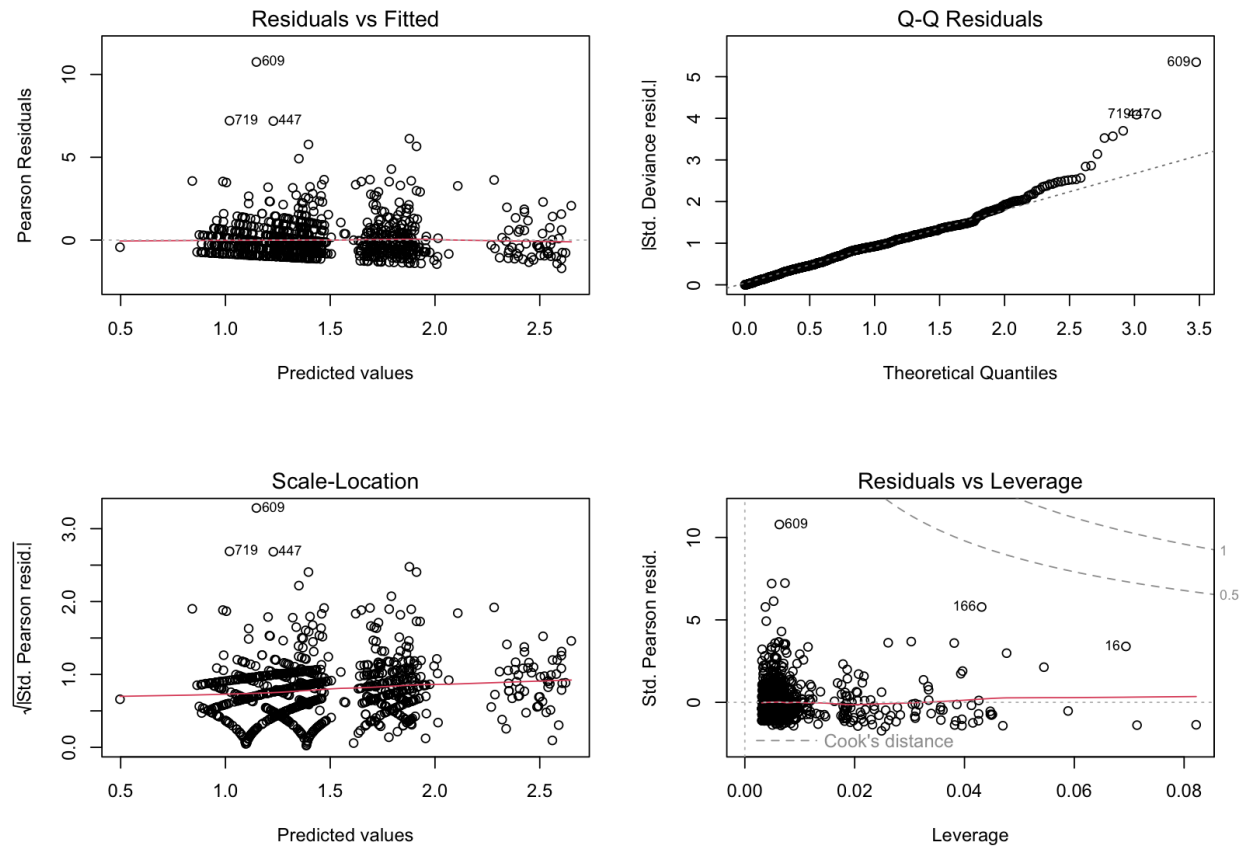


Figure 3: Residual diagnostic plots for the los model

Removing influential points (609, 166, 719) had little effect on coefficients, p-values, or residuals, with a minor AIC improvement. Residual patterns reflect the data's overall structure, including overdispersion, so the points were retained.

Variance Inflation Factor (VIF) analysis showed no significant multicollinearity. The Nega-

tive Binomial regression model is represented as:

$$\begin{aligned} \text{los}_i &\sim \text{Negative Binomial}(\mu_i, \theta) \\ \log(\mu_i) &= \beta_0 + \beta_1(\text{died}) + \beta_2(\text{gender: Male}) \\ &\quad + \beta_3(\text{severity: Moderate}) + \beta_4(\text{severity: Major}) \\ &\quad + \beta_5(\text{severity: Extreme}) + \beta_6(\text{dbp}) \\ &\quad + \beta_7(\text{temp}) \end{aligned} \tag{6}$$

where μ_i is the expected length of stay for observation i , and θ is the dispersion parameter.

3 Results Summary and Interpretation

3.1 Model Mortality

The analyzed model is:

$$\begin{aligned} \widehat{\text{died}}_i &\sim \text{Bernoulli}(\widehat{p}_i) \\ \log\left(\frac{\widehat{p}_i}{1 - \widehat{p}_i}\right) &= \widehat{\beta}_0 + \widehat{\beta}_1(\text{los}) + \widehat{\beta}_2(\text{age}) + \widehat{\beta}_3(\text{gender: Male}) \\ &\quad + \widehat{\beta}_4(\text{BMI}) + \widehat{\beta}_5(\text{spO2}) + \widehat{\beta}_6(\text{respiratory}) + \widehat{\beta}_7(\text{AVPU: Pain}) \\ &\quad + \widehat{\beta}_8(\text{AVPU: Verbal}) + \widehat{\beta}_9(\text{AVPU: Unresponsive}) + \widehat{\beta}_{10}(\text{risk: Moderate}) \\ &\quad + \widehat{\beta}_{11}(\text{risk: Major}) + \widehat{\beta}_{12}(\text{risk: Extreme}) \end{aligned} \tag{7}$$

Table 8: Summary of Logistic Regression Model for Mortality (died)

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	5.19513	5.69439	0.912	0.3616	
los	-0.12198	0.04780	-2.552	0.0107	*
age	0.02404	0.01639	1.467	0.1424	
genderM	0.77324	0.45859	1.686	0.0918	.
bmi	-0.05940	0.03967	-1.497	0.1343	
sp02	-0.12213	0.05649	-2.162	0.0306	*
respiratory	0.07859	0.04575	1.718	0.0858	.
avpuP	1.43040	1.16030	1.233	0.2177	
avpuU	2.05716	0.90623	2.270	0.0232	*
avpuV	-1.38723	1.34656	-1.030	0.3029	
risk2: Moderate	-0.15704	1.03463	-0.152	0.8794	
risk3: Major	1.83559	0.82535	2.224	0.0261	*
risk4: Extreme	4.29719	0.84402	5.091	3.56×10^{-7}	***

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' ' ' '

Table 9: Exponentiated Coefficients (Odds Ratios) from Logistic Regression Model

Variable	Odds Ratio (Exp(Coef))
(Intercept)	180.39
los	0.89
age	1.02
genderM	2.17
bmi	0.94
spO2	0.89
respiratory	1.08
avpuP	4.18
avpuU	7.82
avpuV	0.25
risk2: Moderate	0.85
risk3: Major	6.27
risk4: Extreme	73.49

3.2 Empirical Findings and Interpretation Mortality Model

For **los**, the odds ratio is 0.885, indicating that each additional day reduces mortality odds by 0.89 times. This suggests longer stays may allow effective treatment, though earlier discharges could reflect better initial health.

The odds ratio for **age** is 1.024, meaning each additional year increases mortality odds by 1.02 times, reflecting greater risks for older patients. Male patients (**gender: Male**) have 2.17 times higher mortality odds compared to females, possibly due to biological differences.

bmi has an odds ratio of 0.942, indicating slightly lower mortality odds with higher BMI. Though not significant, this aligns with the "obesity paradox," where moderate obesity may offer protection in critical care.

The odds ratio for **spO2** is 0.885, showing that better oxygenation reduces mortality odds, emphasizing timely oxygen support. Patients with **respiratory** involvement have 1.08 times higher mortality odds, reflecting the impact of respiratory conditions.

For **AVPU**, patients classified as **Pain** have 4.18 times higher mortality odds compared to **Alert**, while **Unresponsive** patients have 7.82 times higher odds. **Verbal** classification lowers mortality odds to 0.25 times, highlighting the role of neurological assessments.

For **risk levels**, **Moderate** risk has an odds ratio of 0.854 (not significant), while **Major** and **Extreme** risks sharply increase mortality odds to 6.27 and 73.49, respectively, underscoring the impact of clinical severity.

3.3 Model Length of Stay

The analysed model can be described as follows:

$$\begin{aligned} \log_i &\sim \text{Negative Binomial}(\widehat{\mu}_i, \widehat{\theta}) \\ \log(\widehat{\mu}_i) &= \widehat{\beta}_0 + \widehat{\beta}_1(\text{died}) + \widehat{\beta}_2(\text{gender: Male}) \\ &\quad + \widehat{\beta}_3(\text{severity: Moderate}) + \widehat{\beta}_4(\text{severity: Major}) \\ &\quad + \widehat{\beta}_5(\text{severity: Extreme}) + \widehat{\beta}_6(\text{dbp}) \\ &\quad + \widehat{\beta}_7(\text{temp}) \end{aligned} \tag{8}$$

Table 10: Coefficients of the Negative Binomial Model for los

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	-2.406971	2.128622	-1.131	0.25815	
died	-0.531563	0.124798	-4.259	2.05e-05	***
genderM	-0.124375	0.044349	-2.804	0.00504	**
severity2 - Moderate	0.281240	0.062386	4.508	6.54e-06	***
severity3 - Major	0.723828	0.065887	10.986	$< 2e - 16$	***
severity4 - Extreme	1.390811	0.088464	15.722	$< 2e - 16$	***
dbp	-0.002537	0.001521	-1.668	0.09540	.
temp	0.037800	0.021704	1.742	0.08157	.

Significance codes: 0 ‘***’, 0.001 ‘**’, 0.01 ‘*’, 0.05 ‘.’, 0.1 ‘ ’

Table 11: Exponentiated Coefficients (Rate Ratios) for the Negative Binomial Model

Variable	Rate Ratio (Exp(Coeff))
(Intercept)	0.0901
died	0.5877
genderM	0.8830
severity2 - Moderate	1.3248
severity3 - Major	2.0623
severity4 - Extreme	4.0181
dbp	0.9975
temp	1.0385

3.4 Empirical Findings and Interpretation Length of Stay

The intercept's rate ratio is 0.0901, representing the expected length of stay when all predictors are at reference levels (`died` = 0, `gender` = Female, `bmi` = 0). This value has limited practical meaning due to the unrealistic baseline scenario.

The rate ratio for `died` is 0.5877, meaning patients who died had 0.59 times the expected length of stay compared to survivors, likely due to rapid deterioration.

For `genderM`, the rate ratio is 0.8830, indicating males had 0.88 times the expected length of stay compared to females, possibly reflecting differences in treatment or health conditions.

Patients with moderate severity had 1.32 times the expected length of stay, major severity 2.06 times, and extreme severity 4.02 times, compared to mild severity. These results indicate a strong positive relationship between severity and length of stay, with more severe conditions requiring longer hospitalization.

The rate ratio for `dbp` is 0.9975, indicating that for each one-unit increase in diastolic blood pressure, the expected length of stay is 0.9975 times as long. This effect is very small and not statistically significant, suggesting minimal impact.

The rate ratio for `temp` is 1.0385, meaning that for each one-unit increase in temperature, the expected length of stay is 1.04 times as long. This suggests a slight increase in hospital stay associated with changes in temperature, reflecting temperature-related conditions.

4 Analysis evaluation

Strengths of the Analysis. The analysis employed a structured methodology. Both mortality and length of stay (LOS) models were refined using stepwise AIC to balance complexity and predictive power. Non-linear relationships were explored with GAMs, and interaction terms were tested, confirming that simpler models were sufficient. The choice of logit for mortality and negative binomial for LOS matched the data characteristics.

Diagnostics were thorough. Multicollinearity was checked with VIF, showing no significant issues. Residuals were carefully analyzed, especially in the LOS model, addressing heteroscedasticity, influential points, and distribution patterns. Adjusting the mortality model threshold demonstrated thoughtful optimization of sensitivity and specificity, supported by detailed performance metrics.

Limitations of the Mortality Model. The primary limitation is the model’s performance with imbalanced data. At the default threshold of 0.5, it missed most deaths, with a low true positive rate (36.4%), despite high overall accuracy. Lowering the threshold to 0.026 improved sensitivity to 84.8% but increased the false positive rate to 17.1%. The small number of death events relative to predictors raises overfitting concerns, reducing generalizability. Calibration was not assessed, leaving uncertainty about the accuracy of predicted probabilities.

Limitations of the LOS Model. While the negative binomial model addressed overdispersion, residual diagnostics suggest unexplained variability. Heteroscedasticity and Q-Q plot deviations indicate struggles with extreme values. Influential points (609, 166, and 719)

suggest potential instability, although their removal did not significantly affect results. Unmeasured factors, such as comorbidities or social determinants, may explain the variability in LOS.

Interconnections and Challenges Across Models. The bidirectional relationship between mortality and LOS complicates interpretation. Short stays may reflect quick recovery or early death, while long stays could indicate complications or intensive care. Neither model fully addresses this interplay. Additionally, time-varying predictors, like oxygen saturation and severity, were treated as static, oversimplifying their dynamic nature.

Addressing Dataset Imbalance. The imbalance (33 deaths vs. 945 survivors) challenges the mortality model. Logistic regression favors the majority class, predicting survival well but missing deaths. Lowering the threshold improved sensitivity but increased false positives, highlighting class separation difficulties. Resampling methods like oversampling, undersampling, or SMOTE could help balance the data and enhance death detection while preserving specificity.

5 Appendix

```
# Load required libraries
library(tidyverse)
library(car)
library(ROCR)
library(pROC)
library(ResourceSelection)
library(MASS)
library(mgcv)
library(ggplot2)

# Load data
hospital <- read.table("hospital.txt", header = TRUE)

# Initial exploration
head(hospital) # View the first few rows of the dataset
table(hospital$died) # Count of deaths and survivors
table(hospital$los) # Distribution of length of stay (los)

# Logistic regression model for mortality
model_mortality <- glm(died ~ ., family = binomial(link = "logit"),
                      data = hospital)

# Model summary
summary(model_mortality)

# Stepwise AIC to remove redundant variables
model_mortality_clean <- stepAIC(model_mortality)
summary(model_mortality_clean)

# Exponentiate coefficients for interpretation
coefficients <- coef(model_mortality_clean)
exp_coefficients <- exp(coefficients)

# Print exponentiated coefficients with labels
for (name in names(exp_coefficients)) {
  cat(name, ":", exp_coefficients[name], "\n")
}

# Check for non-linear relationships using GAM
model_gam <- gam(died ~ s(los) + s(age) + gender + s(bmi) +
                 s(respiratory) + s(spO2) + avpu + risk,
                 family = binomial(link = "logit"),
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        data = hospital)
summary(model_gam)

# Plot GAM results
# Variability in smooth terms indicates returning to stepwise model
par(mfrow = c(1, 1)) # Reset plotting layout
plot(model_gam, shade = TRUE)

# Interaction terms in logistic regression
# Example 1: Interaction between age and severity
model_with_interactions <- glm(died ~ los + age + gender + bmi + sp02 +
                              respiratory + avpu + risk + age:severity,
                              family = binomial(link = "logit"),
                              data = hospital)
summary(model_with_interactions)

# Example 2: Interaction between BMI and risk
model_with_interactions2 <- glm(died ~ los + age + gender + bmi + sp02 +
                              respiratory + avpu + risk + bmi:risk,
                              family = binomial(link = "logit"),
                              data = hospital)
summary(model_with_interactions2)

# Example 3: Interaction between age and risk
model_with_interactions3 <- glm(died ~ los + age + gender + bmi + sp02 +
                              respiratory + avpu + risk + age:risk,
                              family = binomial(link = "logit"),
                              data = hospital)
summary(model_with_interactions3)

# No meaningful results for interactions -> return to model_mortality_clean

# Check multicollinearity with VIF
vif(model_mortality_clean)

# Generate predicted probabilities from the clean model
predicted_probs <- predict(model_mortality_clean, type = "response")

# Evaluate model performance at threshold = 0.5
threshold_1 <- 0.5
predicted_classes_1 <- ifelse(predicted_probs > threshold_1, 1, 0)

# Confusion matrix for threshold = 0.5
conf_matrix_1 <- table(Predicted = predicted_classes_1, Actual = hospital$died)
print("Confusion Matrix (Threshold = 0.5):")

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print(conf_matrix_1)

# Extract metrics from confusion matrix
TP_1 <- conf_matrix_1[2, 2] # True Positives
TN_1 <- conf_matrix_1[1, 1] # True Negatives
FP_1 <- conf_matrix_1[2, 1] # False Positives
FN_1 <- conf_matrix_1[1, 2] # False Negatives

# Calculate accuracy, sensitivity, and false positive rate
accuracy_1 <- (TP_1 + TN_1) / sum(conf_matrix_1)
TPR_1 <- TP_1 / (TP_1 + FN_1) # Sensitivity
FPR_1 <- FP_1 / (FP_1 + TN_1) # False Positive Rate

# Display metrics
cat(sprintf("Threshold 0.5 - Accuracy: %.3f\n", accuracy_1))
cat(sprintf("Threshold 0.5 - Sensitivity: %.3f\n", TPR_1))
cat(sprintf("Threshold 0.5 - False Positive Rate: %.3f\n", FPR_1))

# ROC curve and AUC
roc_curve <- roc(hospital$died, predicted_probs)
auc_value <- auc(roc_curve)

# Plot the ROC curve
plot(
  roc_curve,
  main = "ROC Curve (Threshold = 0.026)", # Main title
  col = "blue", # Color of the ROC curve
  lwd = 2, # Line width
  xlab = "False Positive Rate",
  ylab = "True Positive Rate",
  print.auc = TRUE, # Add AUC to the plot
  print.auc.y = 0.4 # Position of AUC display
)
abline(a = 0, b = 1, col = "gray", lty = 2) # Add diagonal reference line

# Optimal threshold from ROC curve
optimal_threshold <- coords(roc_curve, "best", ret = "threshold")
cat(sprintf("Optimal Threshold: %.3f\n", optimal_threshold))

# Evaluate model at optimal threshold
predicted_classes_optimal <- ifelse(predicted_probs > optimal_threshold, 1, 0)
conf_matrix_optimal <- table(Predicted = predicted_classes_optimal,
                             Actual = hospital$died)

print("Confusion Matrix (Optimal Threshold):")
print(conf_matrix_optimal)

```

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# Metrics for optimal threshold
TP_opt <- conf_matrix_optimal[2, 2] # True Positives
TN_opt <- conf_matrix_optimal[1, 1] # True Negatives
FP_opt <- conf_matrix_optimal[2, 1] # False Positives
FN_opt <- conf_matrix_optimal[1, 2] # False Negatives

accuracy_opt <- (TP_opt + TN_opt) / sum(conf_matrix_optimal)
TPR_opt <- TP_opt / (TP_opt + FN_opt) # Sensitivity
FPR_opt <- FP_opt / (FP_opt + TN_opt) # False Positive Rate

# Display metrics for optimal threshold
cat(sprintf("Optimal Threshold - Accuracy: %.3f\n", accuracy_opt))
cat(sprintf("Optimal Threshold - Sensitivity: %.3f\n", TPR_opt))
cat(sprintf("Optimal Threshold - False Positive Rate: %.3f\n", FPR_opt))

### Length of Stay (LOS) Model ###

# Check mean and variance of response variable
mean_los <- mean(hospital$los)
var_los <- var(hospital$los)
cat(sprintf("Mean LOS: %.2f\n", mean_los))
cat(sprintf("Variance LOS: %.2f\n", var_los))

# Fit Negative Binomial model
model_los <- glm.nb(los ~ ., data = hospital)
summary(model_los)

# Stepwise AIC for Negative Binomial model
model_los_clean <- stepAIC(model_los)
summary(model_los_clean)

# Exponentiate coefficients for interpretation
exp_coefficients <- exp(coef(model_los_clean))
print(exp_coefficients)

# Check for non-linear relationships using GAM
model_los_gam <- gam(los ~ died + gender + severity + s(dbp) + s(temp),
                    family = nb(),
                    data = hospital)
summary(model_los_gam)

# Testing interaction terms for LOS model
interaction_models <- list(
  glm.nb(los ~ died + gender + severity + dbp + temp + severity:gender,

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        data = hospital),
glm.nb(los ~ died + gender + severity + dbp + temp + died:temp,
        data = hospital),
glm.nb(los ~ died + gender + severity + dbp + temp + gender:age,
        data = hospital))

# Iterate over models to display summaries
for (i in seq_along(interaction_models)) {
  cat(sprintf("\nInteraction Model %d Summary:\n", i))
  print(summary(interaction_models[[i]]))
}

# Interaction terms did not improve the model -> return to model_los_clean

# Check multicollinearity
vif(model_los_clean)

# Residual analysis for LOS model
par(mfrow = c(2, 2))
plot(model_los_clean)

# Exclude influential points
hospital_cleaned <- hospital[!rownames(hospital) %in% c("609", "166",
                                                         "719", "447"), ]
model_filtered <- glm.nb(los ~ died + gender + severity + dbp + temp,
                        data = hospital_cleaned)
summary(model_filtered)

par(mfrow = c(2, 2))
plot(model_filtered)

```